TSM 099 **95725**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 2 and (b) x = -2. $x^2 - 2x + 1$	1. (a) 10 pts (b) 10 pts
2. Evaluate the principal square root in each part. $(a)\quad \sqrt{-49}$ $(b)\quad \sqrt{16}$	2. (a) 10 pts (b) 10 pts
3. Evaluate each side at x = 6, -6, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-4}{x-2} = \frac{x^2+5x+6}{x+3}$	3. x = 6 4 pts x = -6 4 pts x = 0 4 pts Conclusion 8 pts
4. Simplify the square root. $\sqrt{18}$	4. 20 pts
5. Reduce each fraction to lowest terms. $(a)\quad \frac{40}{70}$ $(b)\quad \frac{49}{63}$	5. (a) 10 pts (b) 10 pts

95725

TEST PAGE 1

1. (a)

1 — 10 pts

(b)

9 — 10 pts

2. (a)

7i — 10 pts

(b)

4 — 10 pts

3. x = 6

Left 8; right 8; yes — 4 pts

x = -6

Left -4; right -4; yes — 4 pts

x = 0

Left 2; right 2; yes — 4 pts

Conclusion

All tested values work; this alone does not prove an identity. — 8 pts

4.

$3\sqrt{2}$ — 20 pts

5. (a)

$\frac{4}{7}$ — 10 pts

(b)

$\frac{7}{9}$ — 10 pts

TSM 099 **33266**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate the principal square root in each part. $(a)\quad \sqrt{-64}$ $(b)\quad \sqrt{25}$	1. (a) 10 pts (b) 10 pts
2. Reduce each fraction to lowest terms. $(a)\quad \frac{18}{30}$ $(b)\quad \frac{12}{21}$	2. (a) 10 pts (b) 10 pts
3. Evaluate at (a) x = 3 and (b) x = -3. $2(x-1)(x+3)$	3. (a) 10 pts (b) 10 pts
4. Evaluate each side at x = 6, -6, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-4}{x-2} = \frac{x^2+5x+6}{x+3}$	4. x = 6 4 pts x = -6 4 pts x = 0 4 pts Conclusion 8 pts
5. Simplify the square root. $\sqrt{75}$	5. 20 pts

33266

TEST PAGE 1

1. (a)

$8i$ — 10 pts

(b)

5 — 10 pts

2. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

3. (a)

24 — 10 pts

(b)

0 — 10 pts

4. x = 6

Left 8; right 8; yes — 4 pts

x = -6

Left -4; right -4; yes — 4 pts

x = 0

Left 2; right 2; yes — 4 pts

Conclusion

All tested values work; this alone does not prove an identity. — 8 pts

5.

$5\sqrt{3}$ — 20 pts

TSM 099 **99735**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Reduce each fraction to lowest terms. (a) $\frac{24}{42}$ (b) $\frac{14}{21}$	1. (a) 10 pts (b) 10 pts
2. Simplify the square root. $\sqrt{20}$	2. 20 pts
3. Evaluate the principal square root in each part. (a) $\sqrt{-81}$ (b) $\sqrt{36}$	3. (a) 10 pts (b) 10 pts
4. Evaluate each side at x = 6, -6, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-4}{x-2} = \frac{x^2+5x+6}{x+3}$	4. x = 6 4 pts x = -6 4 pts x = 0 4 pts Conclusion 8 pts
5. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$	5. 20 pts

99735

TEST PAGE 1

1. (a)

$\frac{4}{7}$ — 10 pts

(b)

$\frac{2}{3}$ — 10 pts

2.

$2\sqrt{5}$

20 pts

3. (a)

9i — 10 pts

(b)

6 — 10 pts

4. x = 6

Left 8; right 8; yes — 4 pts

x = -6

Left -4; right -4; yes — 4 pts

x = 0

Left 2; right 2; yes — 4 pts

Conclusion

All tested values work; this alone does not prove an identity. — 8 pts

5.

$x = \frac{1}{2},\ 4$

20 pts

TSM 099 **68448**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{18}$	1. 20 pts
2. Evaluate the principal square root in each part. (a) $\sqrt{49}$ (b) $\sqrt{-100}$	2. (a) 10 pts (b) 10 pts
3. Reduce each fraction to lowest terms. (a) $\frac{30}{48}$ (b) $\frac{14}{21}$	3. (a) 10 pts (b) 10 pts
4. Evaluate at (a) x = 2 and (b) x = -2. $2x^2 - 2x + 4$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 7, -7, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-9}{x-3} = \frac{2x^2+11x+12}{x+4}$	5. x = 7 4 pts x = -7 4 pts x = 0 4 pts Conclusion 8 pts

68448

TEST PAGE 1

1.

$3\sqrt{2}$

20 pts

2. (a)

7

10 pts

(b)

10i

10 pts

3. (a)

$\frac{5}{8}$

10 pts

(b)

$\frac{2}{3}$

10 pts

4. (a)

8

10 pts

(b)

16

10 pts

5. x = 7

Left 10; right 17; no

4 pts

x = -7

Left -4; right -11; no

4 pts

x = 0

Left 3; right 3; yes

4 pts

Conclusion

Only the listed matching value works; the equation is not an identity.

8 pts

TSM 099 **29055**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{75}$	1. 20 pts
2. Reduce each fraction to lowest terms. (a) $\frac{14}{18}$ (b) $\frac{70}{112}$	2. (a) 10 pts (b) 10 pts
3. Evaluate at (a) x = 3 and (b) x = -3. $(x-1)(x+2)$	3. (a) 10 pts (b) 10 pts
4. Evaluate the principal square root in each part. (a) $\sqrt{-121}$ (b) $\sqrt{64}$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 7, -7, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-9}{x-3} = \frac{2x^2+11x+12}{x+4}$	5. x = 7 4 pts x = -7 4 pts x = 0 4 pts Conclusion 8 pts

29055

TEST PAGE 1

1.

$5\sqrt{3}$

20 pts

2. (a)

$\frac{7}{9}$

10 pts

(b)

$\frac{5}{8}$

10 pts

3. (a)

10

10 pts

(b)

4

10 pts

4. (a)

11i

10 pts

(b)

8

10 pts

5. x = 7

Left 10; right 17; no

4 pts

x = -7

Left -4; right -11; no

4 pts

x = 0

Left 3; right 3; yes

4 pts

Conclusion

Only the listed matching value works; the equation is not an identity.

8 pts

TSM 099 **63742**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate the principal square root in each part. $(a)\quad \sqrt{-25}$ $(b)\quad \sqrt{81}$	1. (a) 10 pts (b) 10 pts
2. Evaluate each side at x = 7, -7, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-9}{x-3} = \frac{2x^2+11x+12}{x+4}$	2. x = 7 4 pts x = -7 4 pts x = 0 4 pts Conclusion 8 pts
3. Simplify the square root. $\sqrt{20}$	3. 20 pts
4. Reduce each fraction to lowest terms. $(a)\quad \frac{18}{30}$ $(b)\quad \frac{8}{14}$	4. (a) 10 pts (b) 10 pts
5. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$	5. 20 pts

63742

TEST PAGE 1

1. (a)

5i — 10 pts

(b)

9 — 10 pts

2. x = 7

Left 10; right 17; no — 4 pts

x = -7

Left -4; right -11; no — 4 pts

x = 0

Left 3; right 3; yes — 4 pts

Conclusion

Only the listed matching value works; the equation is not an identity. — 8 pts

3.

$2\sqrt{5}$

20 pts

4. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

5.

$x = \frac{1}{2},\ 4$

20 pts

TSM 099 **87039**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{18}$	1. 20 pts
2. Reduce each fraction to lowest terms. $(a)\quad \frac{10}{15}$ $(b)\quad \frac{18}{30}$	2. (a) 10 pts (b) 10 pts
3. Evaluate at (a) x = 2 and (b) x = -2. $x^2 - 2x + 3$	3. (a) 10 pts (b) 10 pts
4. Evaluate the principal square root in each part. $(a)\quad \sqrt{100}$ $(b)\quad \sqrt{-36}$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 8, -8, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-16}{x-4} = \frac{x^2+10x+25}{x+5}$	5. x = 8 4 pts x = -8 4 pts x = 0 4 pts Conclusion 8 pts

87039

TEST PAGE 1

1.

$3\sqrt{2}$

20 pts

2. (a)

$\frac{2}{3}$

10 pts

(b)

$\frac{3}{5}$

10 pts

3. (a)

3

10 pts

(b)

11

10 pts

4. (a)

10

10 pts

(b)

6i

10 pts

5. x = 8

Left 12; right 13; no

4 pts

x = -8

Left -4; right -3; no

4 pts

x = 0

Left 4; right 5; no

4 pts

Conclusion

None of these values works; this alone does not prove no solution.

8 pts

TSM 099 **30359**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 8, -8, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-16}{x-4} = \frac{x^2+10x+25}{x+5}$	1. x = 8 4 pts x = -8 4 pts x = 0 4 pts Conclusion 8 pts
2. Simplify the square root. $\sqrt{75}$	2. 20 pts
3. Reduce each fraction to lowest terms. (a) $\frac{6}{10}$ (b) $\frac{42}{70}$	3. (a) 10 pts (b) 10 pts
4. Evaluate at (a) x = 3 and (b) x = -3. $2(x-1)(x+3)$	4. (a) 10 pts (b) 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{-49}$ (b) $\sqrt{16}$	5. (a) 10 pts (b) 10 pts

30359

TEST PAGE 1

1. x = 8

Left 12; right 13; no — 4 pts

x = -8

Left -4; right -3; no — 4 pts

x = 0

Left 4; right 5; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

2.

$5\sqrt{3}$ — 20 pts

3. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{3}{5}$ — 10 pts

4. (a)

24 — 10 pts

(b)

0 — 10 pts

5. (a)

7i — 10 pts

(b)

4 — 10 pts

TSM 099 **35974**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{20}$	1. 20 pts
2. Evaluate the principal square root in each part. $(a)\quad \sqrt{-64}$ $(b)\quad \sqrt{25}$	2. (a) 10 pts (b) 10 pts
3. Evaluate each side at x = 8, -8, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-16}{x-4} = \frac{x^2+10x+25}{x+5}$	3. x = 8 4 pts x = -8 4 pts x = 0 4 pts Conclusion 8 pts
4. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$	4. 20 pts
5. Reduce each fraction to lowest terms. $(a)\quad \frac{15}{25}$ $(b)\quad \frac{12}{18}$	5. (a) 10 pts (b) 10 pts

35974

TEST PAGE 1

1.

$2\sqrt{5}$

20 pts

2. (a)

8i

10 pts

(b)

5

10 pts

3. x = 8

Left 12; right 13; no

4 pts

x = -8

Left -4; right -3; no

4 pts

x = 0

Left 4; right 5; no

4 pts

Conclusion

None of these values works; this alone does not prove no solution.

8 pts

4.

$x = \frac{1}{2},\ 4$

20 pts

5. (a)

$\frac{3}{5}$

10 pts

(b)

$\frac{2}{3}$

10 pts

TSM 099 **59750**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{18}$	1. 20 pts
2. Reduce each fraction to lowest terms. (a) $\frac{25}{40}$ (b) $\frac{20}{30}$	2. (a) 10 pts (b) 10 pts
3. Evaluate each side at x = 9, -9, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-25}{x-5} = \frac{x^2+12x+36}{x+6}$	3. x = 9 4 pts x = -9 4 pts x = 0 4 pts Conclusion 8 pts
4. Evaluate the principal square root in each part. (a) $\sqrt{36}$ (b) $\sqrt{-81}$	4. (a) 10 pts (b) 10 pts
5. Evaluate at (a) x = 2 and (b) x = -2. $2x^2 - 2x + 2$	5. (a) 10 pts (b) 10 pts

59750

TEST PAGE 1

1.

$3\sqrt{2}$

20 pts

2. (a)

$\frac{5}{8}$

10 pts

(b)

$\frac{2}{3}$

10 pts

3. x = 9

Left 14; right 15; no

4 pts

x = -9

Left -4; right -3; no

4 pts

x = 0

Left 5; right 6; no

4 pts

Conclusion

None of these values works; this alone does not prove no solution.

8 pts

4. (a)

6

10 pts

(b)

9i

10 pts

5. (a)

6

10 pts

(b)

14

10 pts

TSM 099 **79589**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{75}$	1. 20 pts
2. Evaluate the principal square root in each part. $(a)\quad \sqrt{-100}$ $(b)\quad \sqrt{49}$	2. (a) 10 pts (b) 10 pts
3. Reduce each fraction to lowest terms. $(a)\quad \frac{35}{56}$ $(b)\quad \frac{70}{112}$	3. (a) 10 pts (b) 10 pts
4. Evaluate at (a) x = 3 and (b) x = -3. $(x-1)(x+2)$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 9, -9, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-25}{x-5} = \frac{x^2+12x+36}{x+6}$	5. x = 9 4 pts x = -9 4 pts x = 0 4 pts Conclusion 8 pts

79589

TEST PAGE 1

1.

$5\sqrt{3}$

20 pts

2. (a)

10i

10 pts

(b)

7

10 pts

3. (a)

$\frac{5}{8}$

10 pts

(b)

$\frac{5}{8}$

10 pts

4. (a)

10

10 pts

(b)

4

10 pts

5. x = 9

Left 14; right 15; no

4 pts

x = -9

Left -4; right -3; no

4 pts

x = 0

Left 5; right 6; no

4 pts

Conclusion

None of these values works; this alone does not prove no solution.

8 pts

TSM 099 **57635**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate the principal square root in each part. $(a)\quad \sqrt{-121}$ $(b)\quad \sqrt{64}$	1. (a) 10 pts (b) 10 pts
2. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	2. 20 pts
3. Simplify the square root. $\sqrt{20}$	3. 20 pts
4. Reduce each fraction to lowest terms. $(a)\quad \frac{15}{24}$ $(b)\quad \frac{18}{30}$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 9, -9, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2 - 25}{x - 5} = \frac{x^2 + 12x + 36}{x + 6}$	5. x = 9 4 pts x = -9 4 pts x = 0 4 pts Conclusion 8 pts

57635

TEST PAGE 1

1. (a)

11i — 10 pts

(b)

8 — 10 pts

2.

$x = \frac{1}{2},\ 4$

20 pts

3.

$2\sqrt{5}$

20 pts

4. (a)

$\frac{5}{8}$ — 10 pts

(b)

$\frac{3}{5}$ — 10 pts

5. x = 9

Left 14; right 15; no — 4 pts

x = -9

Left -4; right -3; no — 4 pts

x = 0

Left 5; right 6; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

TSM 099 **58938**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 10, -10, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-36}{x-6} = \frac{x^2+14x+49}{x+7}$	1. x = 10 — 4 pts x = -10 — 4 pts x = 0 — 4 pts Conclusion — 8 pts
2. Simplify the square root. $\sqrt{18}$	2. — 20 pts
3. Reduce each fraction to lowest terms. (a) $\frac{60}{105}$ (b) $\frac{15}{25}$	3. (a) — 10 pts (b) — 10 pts
4. Evaluate at (a) x = 2 and (b) x = -2. x^2-2x+1	4. (a) — 10 pts (b) — 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{81}$ (b) $\sqrt{-25}$	5. (a) — 10 pts (b) — 10 pts

58938

TEST PAGE 1

1. x = 10

Left 16; right 17; no — 4 pts

x = -10

Left -4; right -3; no — 4 pts

x = 0

Left 6; right 7; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

---

2.

$3\sqrt{2}$

20 pts

---

3. (a)

$\frac{4}{7}$ — 10 pts

(b)

$\frac{3}{5}$ — 10 pts

---

4. (a)

1 — 10 pts

(b)

9 — 10 pts

---

5. (a)

9 — 10 pts

(b)

5i — 10 pts

TSM 099 **99854**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 3 and (b) x = -3. $2(x-1)(x+3)$	1. (a) 10 pts (b) 10 pts
2. Simplify the square root. $\sqrt{75}$	2. 20 pts
3. Reduce each fraction to lowest terms. $(a)\ \frac{30}{45}$ $(b)\ \frac{14}{18}$	3. (a) 10 pts (b) 10 pts
4. Evaluate the principal square root in each part. $(a)\ \sqrt{100}$ $(b)\ \sqrt{-36}$	4. (a) 10 pts (b) 10 pts
5. Evaluate each side at x = 10, -10, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-36}{x-6} = \frac{x^2+14x+49}{x+7}$	5. x = 10 4 pts x = -10 4 pts x = 0 4 pts Conclusion 8 pts

99854

TEST PAGE 1

1. (a)

24 — 10 pts

(b)

0 — 10 pts

2.

$5\sqrt{3}$ — 20 pts

3. (a)

$\frac{2}{3}$ — 10 pts

(b)

$\frac{7}{9}$ — 10 pts

4. (a)

10 — 10 pts

(b)

6i — 10 pts

5. x = 10

Left 16; right 17; no — 4 pts

x = -10

Left -4; right -3; no — 4 pts

x = 0

Left 6; right 7; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

TSM 099 **68237**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 10, -10, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-36}{x-6} = \frac{x^2+14x+49}{x+7}$	1. x = 10 — 4 pts x = -10 — 4 pts x = 0 — 4 pts Conclusion — 8 pts
2. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$	2. — 20 pts
3. Simplify the square root. $\sqrt{20}$	3. — 20 pts
4. Reduce each fraction to lowest terms. $(a)\ \frac{42}{54}$ $(b)\ \frac{20}{35}$	4. (a) — 10 pts (b) — 10 pts
5. Evaluate the principal square root in each part. $(a)\ \sqrt{-49}$ $(b)\ \sqrt{16}$	5. (a) — 10 pts (b) — 10 pts

68237

TEST PAGE 1

1. x = 10

Left 16; right 17; no — 4 pts

x = -10

Left -4; right -3; no — 4 pts

x = 0

Left 6; right 7; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

2.

$x = \frac{1}{2},\ 4$

20 pts

3.

$2\sqrt{5}$

20 pts

4. (a)

$\frac{7}{9}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

5. (a)

7i — 10 pts

(b)

4 — 10 pts

TSM 099 **61862**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 2 and (b) x = -2. $2x^2 - 2x + 4$	1. (a) 10 pts (b) 10 pts
2. Evaluate each side at x = 11, -11, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2 - 49}{x - 7} = \frac{x^2 + 16x + 64}{x + 8}$	2. x = 11 4 pts x = -11 4 pts x = 0 4 pts Conclusion 8 pts
3. Reduce each fraction to lowest terms. (a) $\frac{15}{25}$ (b) $\frac{40}{70}$	3. (a) 10 pts (b) 10 pts
4. Evaluate the principal square root in each part. (a) $\sqrt{-64}$ (b) $\sqrt{25}$	4. (a) 10 pts (b) 10 pts
5. Simplify the square root. $\sqrt{18}$	5. 20 pts

61862

TEST PAGE 1

1. (a)

8 — 10 pts

(b)

16 — 10 pts

2. x = 11

Left 18; right 19; no — 4 pts

x = -11

Left -4; right -3; no — 4 pts

x = 0

Left 7; right 8; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

3. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

4. (a)

8i — 10 pts

(b)

5 — 10 pts

5.

$3\sqrt{2}$

20 pts

TSM 099 **90851**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 3 and (b) x = -3. $(x-1)(x+2)$	1. (a) 10 pts (b) 10 pts
2. Evaluate the principal square root in each part. $(a)\quad \sqrt{-81}$ $(b)\quad \sqrt{36}$	2. (a) 10 pts (b) 10 pts
3. Evaluate each side at x = 11, -11, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-49}{x-7} = \frac{x^2+16x+64}{x+8}$	3. x = 11 4 pts x = -11 4 pts x = 0 4 pts Conclusion 8 pts
4. Simplify the square root. $\sqrt{75}$	4. 20 pts
5. Reduce each fraction to lowest terms. $(a)\quad \frac{70}{112}$ $(b)\quad \frac{49}{63}$	5. (a) 10 pts (b) 10 pts

90851

TEST PAGE 1

1. (a)

10 — 10 pts

(b)

4 — 10 pts

2. (a)

9i — 10 pts

(b)

6 — 10 pts

3. x = 11

Left 18; right 19; no — 4 pts

x = -11

Left -4; right -3; no — 4 pts

x = 0

Left 7; right 8; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

4.

$5\sqrt{3}$ — 20 pts

5. (a)

$\frac{5}{8}$ — 10 pts

(b)

$\frac{7}{9}$ — 10 pts

TSM 099 **47886**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate the principal square root in each part. $(a)\quad \sqrt{49}$ $(b)\quad \sqrt{-100}$	1. (a) 10 pts (b) 10 pts
2. Simplify the square root. $\sqrt{20}$	2. 20 pts
3. Reduce each fraction to lowest terms. $(a)\quad \frac{45}{75}$ $(b)\quad \frac{8}{14}$	3. (a) 10 pts (b) 10 pts
4. Evaluate each side at x = 11, -11, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-49}{x-7} = \frac{x^2+16x+64}{x+8}$	4. x = 11 4 pts x = -11 4 pts x = 0 4 pts Conclusion 8 pts
5. Evaluate both values in the formula for a = 2, b = -9, c = 4. $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$	5. 20 pts

47886

TEST PAGE 1

1. (a)

7 — 10 pts

(b)

10i — 10 pts

2.

$2\sqrt{5}$

20 pts

3. (a)

$\frac{3}{5}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

4. x = 11

Left 18; right 19; no — 4 pts

x = -11

Left -4; right -3; no — 4 pts

x = 0

Left 7; right 8; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

5.

$x = \frac{1}{2},\ 4$

20 pts

TSM 099 **71533**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 12, -12, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-64}{x-8} = \frac{x^2+18x+81}{x+9}$	1. x = 12 — 4 pts x = -12 — 4 pts x = 0 — 4 pts Conclusion — 8 pts
2. Evaluate at (a) x = 2 and (b) x = -2. x^2-2x+3	2. (a) — 10 pts (b) — 10 pts
3. Reduce each fraction to lowest terms. (a) $\frac{8}{14}$ (b) $\frac{24}{42}$	3. (a) — 10 pts (b) — 10 pts
4. Simplify the square root. $\sqrt{18}$	4. — 20 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{-121}$ (b) $\sqrt{64}$	5. (a) — 10 pts (b) — 10 pts

71533

TEST PAGE 1

1. x = 12

Left 20; right 21; no — 4 pts

x = -12

Left -4; right -3; no — 4 pts

x = 0

Left 8; right 9; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

2. (a)

3 — 10 pts

(b)

11 — 10 pts

3. (a)

$\frac{4}{7}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

4.

$3\sqrt{2}$

20 pts

5. (a)

11i — 10 pts

(b)

8 — 10 pts

TSM 099 **34863**

Arithmetic Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 12, -12, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-64}{x-8} = \frac{x^2+18x+81}{x+9}$	1. x = 12 — 4 pts x = -12 — 4 pts x = 0 — 4 pts Conclusion — 8 pts
2. Reduce each fraction to lowest terms. (a) $\frac{25}{40}$ (b) $\frac{60}{105}$	2. (a) — 10 pts (b) — 10 pts
3. Simplify the square root. $\sqrt{75}$	3. — 20 pts
4. Evaluate at (a) x = 3 and (b) x = -3. $2(x-1)(x+3)$	4. (a) — 10 pts (b) — 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{-25}$ (b) $\sqrt{81}$	5. (a) — 10 pts (b) — 10 pts

34863

TEST PAGE 1

1. x = 12

Left 20; right 21; no — 4 pts

x = -12

Left -4; right -3; no — 4 pts

x = 0

Left 8; right 9; no — 4 pts

Conclusion

None of these values works; this alone does not prove no solution. — 8 pts

2. (a)

$\frac{5}{8}$ — 10 pts

(b)

$\frac{4}{7}$ — 10 pts

3.

$5\sqrt{3}$ — 20 pts

4. (a)

24 — 10 pts

(b)

0 — 10 pts

5. (a)

5i — 10 pts

(b)

9 — 10 pts